

hw 7. P303.  $T_1$  (1)(2)(3)  $T_2$  (2)(4)(6)

$T_1$  (1) 由于  $x$  的取值为  $-\infty < x < +\infty$ , 故可对  $x$  作傅里叶变换。记  $\bar{u}(\lambda, y) = \mathcal{F}\{u(x, y)\}$

$\bar{f}(\lambda) = \mathcal{F}\{f(x)\}$ , 由此可得下式:

$$\begin{cases} \frac{d^2 \bar{u}}{dy^2} - \lambda^2 \bar{u} = 0 \\ \bar{u}(\lambda, 0) = \bar{f}(\lambda) \end{cases} \quad \text{解得: } \bar{u}(\lambda, y) = A(\lambda)e^{|\lambda|y} + B(\lambda)e^{-|\lambda|y}$$

代入边界条件即可得:  $\bar{u}(\lambda, 0) = A(\lambda) + B(\lambda) = \bar{f}(\lambda)$

又当  $x^2 + y^2 \rightarrow +\infty$  时,  $u(x, y) \rightarrow 0$  故  $A(\lambda) = 0$

故  $\bar{u}(\lambda, y) = \bar{f}(\lambda) e^{-|\lambda|y}$

$$\begin{aligned} u(x, y) &= \mathcal{F}^{-1}\{\bar{u}(\lambda, y)\} = \mathcal{F}^{-1}\{\bar{f}(\lambda)\} * \mathcal{F}^{-1}\{e^{-|\lambda|y}\} \\ &= f(x) * \frac{y}{\pi(x^2 + y^2)} = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{y f(\xi)}{(x - \xi)^2 + y^2} d\xi \end{aligned}$$

$$\begin{aligned} \text{其中: } \mathcal{F}^{-1}\{e^{-|\lambda|y}\} &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-|\lambda|y} e^{i\lambda x} d\lambda \\ &= \frac{1}{\pi} \int_0^{+\infty} e^{-\lambda y} \cos \lambda x d\lambda = \frac{1}{\pi} \frac{y}{x^2 + y^2} \end{aligned}$$

(2)  $\bar{u}(t, \lambda) = \mathcal{F}\{u(t, x)\}$  记  $\bar{f}(t, \lambda) = \mathcal{F}\{f(t, x)\}$

$$\text{则方程如下: } \begin{cases} \frac{d\bar{u}}{dt} + a^2 \lambda^2 \bar{u} = \bar{f}(t, \lambda) \\ \bar{u}(0, \lambda) = 0 \end{cases}$$

得到通解为:

$$\bar{u}(t, \lambda) = e^{-a^2 \lambda^2 t} \int_0^t e^{a^2 \lambda^2 \tau} \bar{f}(\tau, \lambda) d\tau$$

对其作反变换得:

$$\begin{aligned} u(t, x) &= \mathcal{F}^{-1}\{\bar{u}(t, \lambda)\} = \int_0^t \mathcal{F}^{-1}\{e^{-a^2 \lambda^2 (t-\tau)}\} * \mathcal{F}^{-1}\{\bar{f}(\tau, \lambda)\} d\tau \\ &= \int_0^t \frac{e^{-\frac{x^2}{4a^2(t-\tau)}}}{2a\sqrt{\pi(t-\tau)}} * f(\tau, x) d\tau \end{aligned}$$

$$= \frac{1}{2a\sqrt{\pi}} \int_0^t \int_{-\infty}^{+\infty} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \frac{f(\tau, \xi)}{\sqrt{t-\tau}} d\xi d\tau$$

利用公式: (书P294)  $F^{-1}\{e^{-a^2\lambda^2 t}\} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a^2\lambda^2 t} e^{-i\lambda x} d\lambda$   
 $= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-a^2\lambda^2 t} \cos \lambda x d\lambda = \frac{1}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}}$

(3) 由于  $x$  的取值范围为  $0 < x < +\infty$ , 故对  $x$  作正弦变换:

记  $\bar{u}(t, \lambda) = \int_0^{+\infty} u(t, x) \sin \lambda x dx$

$$\begin{aligned} F_s\left\{\frac{\partial^2 u}{\partial x^2}\right\} &= \int_0^{+\infty} \frac{\partial^2 u}{\partial x^2} \sin \lambda x dx = \frac{\partial u}{\partial x} \sin \lambda x \Big|_0^{+\infty} - \lambda \int_0^{+\infty} \frac{\partial u}{\partial x} \cos \lambda x dx \\ &= -\lambda u \cos \lambda x \Big|_0^{+\infty} - \lambda^2 \int_0^{+\infty} u \sin \lambda x dx = \lambda \varphi(t) - \lambda^2 \bar{u} \end{aligned}$$

故可得如下方程:

$$\begin{cases} \frac{d\bar{u}}{dt} + a^2 \lambda^2 \bar{u} = a^2 \lambda \varphi(t) \\ \bar{u}(0, \lambda) = 0 \end{cases}$$

其通解为:  $\bar{u}(t, \lambda) = e^{-a^2 \lambda^2 t} \int_0^t e^{a^2 \lambda^2 \tau} a^2 \lambda \varphi(\tau) d\tau = \int_0^t e^{a^2 \lambda^2 (\tau-t)} a^2 \lambda \varphi(\tau) d\tau$

因此作反变换得到:

$$\begin{aligned} u(t, x) &= \frac{2}{\pi} \int_0^t a^2 \varphi(\tau) d\tau \int_0^{+\infty} \lambda (\sin \lambda x) e^{-a^2 \lambda^2 (t-\tau)} d\lambda \\ &= \frac{x}{2a\sqrt{\pi}} \int_0^t (t-\tau)^{-\frac{3}{2}} \varphi(\tau) e^{-\frac{x^2}{4a^2(t-\tau)}} d\tau \end{aligned}$$

T2. (2) 对  $t$  作拉普拉斯变换.  $\tilde{u}(p, x) = \mathcal{L}\{u(t, x)\}$

得到如下方程:

$$\begin{cases} p\tilde{u} - u_1 = a^2 \tilde{u}_{xx} \\ \tilde{u}_x(p, 0) = 0, \quad \tilde{u}(p, l) = \frac{u_0}{p} \end{cases}$$

通解为:  $\tilde{u}(p, x) = \frac{u_1}{p} + A \operatorname{sh} \frac{\sqrt{p}}{a} x + B \operatorname{ch} \frac{\sqrt{p}}{a} x$

代入边界条件有:

$$\begin{cases} \tilde{u}_x(p, 0) = A \cdot \frac{\sqrt{p}}{a} = 0 \Rightarrow A = 0 \\ \tilde{u}(p, l) = \frac{u_1}{p} + B \operatorname{ch} \frac{\sqrt{p}}{a} l = \frac{u_0}{p} \Rightarrow B = \frac{u_0 - u_1}{p \operatorname{ch} \frac{\sqrt{p}}{a} l} \\ \Rightarrow \tilde{u}(p, x) = \frac{u_1}{p} + \frac{u_0 - u_1}{p} \frac{\operatorname{ch} \frac{\sqrt{p}}{a} x}{\operatorname{ch} \frac{\sqrt{p}}{a} l} \end{cases}$$

$$u(t, x) = \mathcal{L}^{-1} \{ \tilde{u}(p, x) \} = u_1 + (u_0 - u_1) \left[ 1 + \frac{4}{\pi} \sum_{k=0}^{+\infty} \frac{(-1)^{k+1}}{2k+1} \cos \frac{(2k+1)\pi x}{2l} \cdot e^{-\left[ \frac{(2k+1)\pi a}{2l} \right]^2 t} \right]$$

(4)  $\tilde{u}(p, x) = \mathcal{L} \{ u(t, x) \}$  得到如下方程:

$$\begin{cases} p^2 \tilde{u} - b = a^2 \tilde{u}_{xx} \\ \tilde{u}(p, 0) = 0, \lim_{x \rightarrow +\infty} \tilde{u}_x = 0 \end{cases} \quad \text{通解为: } \tilde{u}(p, x) = A e^{\frac{p}{a} x} + B e^{-\frac{p}{a} x} + \frac{b}{p^2}$$

代入边界条件有:  $\begin{cases} \tilde{u}(p, 0) = A + B + \frac{b}{p^2} = 0 \\ \lim_{x \rightarrow +\infty} \tilde{u}_x = \lim_{x \rightarrow +\infty} (A e^{\frac{p}{a} x} + B e^{-\frac{p}{a} x}) = 0 \end{cases}$

$$\Rightarrow \begin{cases} A = 0 \\ B = -\frac{b}{p^2} \end{cases} \quad \text{故 } \tilde{u}(p, x) = \frac{b}{p^2} (1 - e^{-\frac{p}{a} x})$$

$$u(t, x) = \mathcal{L}^{-1} \{ \tilde{u}(p, x) \} = b t - b \left( t - \frac{x}{a} \right) h \left( t - \frac{x}{a} \right)$$

$h(t)$  为单位阶数.

(6)  $\tilde{u}(p, x) = \mathcal{L} \{ u(t, x) \}$  得到如下方程:

$$\begin{cases} p \tilde{u} - 4 \cos 5\pi x + 2 \cos 3\pi x = \tilde{u}_{xx} \\ \tilde{u}_x(p, 0) = 0, \tilde{u}_x(p, 2) = 0 \end{cases} \quad \text{设特解为 } \tilde{u}^*(p, x) = C \cos 5\pi x + D \cos 3\pi x$$

代入原方程得:  $C = \frac{4}{p + \pi^2}, D = -\frac{2}{p + 9\pi^2}$

通解为:  $\tilde{u}(p, x) = A e^{\sqrt{p} x} + B e^{-\sqrt{p} x} + \frac{4}{p + \pi^2} \cos 5\pi x - \frac{2}{p + 9\pi^2} \cos 3\pi x$

代入边界条件有:  $\begin{cases} \tilde{u}_x(p, 0) = \sqrt{p}(A - B) = 0 \\ \tilde{u}_x(p, 2) = \sqrt{p}(A e^{2\sqrt{p}} + B e^{-2\sqrt{p}}) = 0 \end{cases} \Rightarrow \begin{cases} A = 0 \\ B = 0 \end{cases}$

$$\Rightarrow \tilde{u}(p, x) = \frac{4 \cos \pi x}{p + \pi^2} - \frac{2 \cos 3\pi x}{p + 9\pi^2}$$

$$\text{故 } u(t, x) = \mathcal{L}^{-1} \{ \tilde{u}(p, x) \} = 4 e^{-\pi^2 t} \cos \pi x - 2 e^{-9\pi^2 t} \cos 3\pi x$$